

Devops day6-Prometheus grafana

Prometheus is an open-source system monitoring and alerting toolkit originally built at SoundCloud. It is now a standalone open source project . Prometheus joined the Cloud Native Computing Foundation in 2016 as the second hosted project, after Kubernetes.

Features,

1. a multi-dimensional data model with time series data identified by metric name and key/value pairs
2. PromQL, a flexible query language to leverage this dimensionality
3. no reliance on distributed storage; single server nodes are autonomous
4. time series collection happens via a pull model over HTTP
5. pushing time series is supported via an intermediary gateway
6. targets are discovered via service discovery or static configuration
7. multiple modes of graphing and dashboarding support

Prometheus Installation:

Username Creation:

```
sudo useradd \  
--system \  
--no-create-home \  
--shell /bin/false Prometheus
```

Commands:

```
wget  
https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-  
2.47.1.linux-amd64.tar.gz  
  
tar -xvf prometheus-2.47.1.linux-amd64.tar.gz  
  
sudo mkdir -p /data /etc/prometheus  
  
cd prometheus-2.47.1.linux-amd64/  
  
sudo mv prometheus promtool /usr/local/bin/
```

```

sudo mv consoles/ console_libraries/ /etc/prometheus/

sudo mv prometheus.yml /etc/prometheus/prometheus.yml

sudo chown -R prometheus:prometheus /etc/prometheus/ /data/

cd

rm -rf prometheus-2.47.1.linux-amd64.tar.gz

prometheus --version

sudo vim /etc/systemd/system/prometheus.service

```

```

pavis@pavish-kumar: ~
^C Stopping tunnel for service my-service.
pavis@pavish-kumar:~$ sudo useradd \
--system \
--no-create-home \
--shell /bin/false prometheus
[sudo] password for pavis:
pavis@pavish-kumar:~$ wget https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-2.47.1.linux-amd64.tar.gz
tar -xvf prometheus-2.47.1.linux-amd64.tar.gz
sudo mkdir -p /data /etc/prometheus
cd prometheus-2.47.1.linux-amd64/
sudo mv prometheus promtool /usr/local/bin/
sudo mv consoles/ console_libraries/ /etc/prometheus/
sudo mv prometheus.yml /etc/prometheus/prometheus.yml
sudo chown -R prometheus:prometheus /etc/prometheus/ /data/
--2025-03-22 06:29:45-- https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-2.47.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com):20.207.73.82:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/6838921/2f9b7b37-63a0-428b-adb5-0294482fd743?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250322%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250322T062945Z&X-Amz-Expires=300&X-Amz-Signature=c3d27259d96a4e0aace6f7560a29265b5df95462136caa5c84598a07b965249c&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-03-22 06:29:46-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/6838921/2f9b7b37-63a0-428b-adb5-0294482fd743?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=releaseassetproduction%2F20250322%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250322T062945Z&X-Amz-Expires=300&X-Amz-Signature=c3d27259d96a4e0aace6f7560a29265b5df95462136caa5c84598a07b965249c&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com):185.199.108.133:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 95713066 (91M) [application/octet-stream]
Saving to: 'prometheus-2.47.1.linux-amd64.tar.gz'

pavis@pavish-kumar: ~
prometheus-2.47.1.linux-amd64/prometheus
pavis@pavish-kumar:~/prometheus-2.47.1.linux-amd64$ cd
rm -rf prometheus-2.47.1.linux-amd64.tar.gz
prometheus --version
sudo vim /etc/systemd/system/prometheus.service
prometheus, version 2.47.1 (branch: HEAD, revision: c4d1a8beff37cc004f1dc4ab9d2e73193f51aaeb)
  build user:   root@4829330363be
  build date:   20231004-10:31:16
  go version:   go1.21.1
  platform:     linux/amd64
  tags:         netgo, builtinassets,stringlabels
pavis@pavish-kumar:~$ sudo vim /etc/systemd/system/prometheus.service
pavis@pavish-kumar:~$ sudo vim /etc/systemd/system/prometheus.service
pavis@pavish-kumar:~$ sudo systemctl enable prometheus
sudo systemctl start prometheus
sudo systemctl status prometheus
journalctl -u prometheus -f --no-pager
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
● prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-03-22 06:40:12 UTC; 21ms ago
     Main PID: 14349 (prometheus)
       Tasks: 1 (limit: 4592)
      Memory: 2.4M
    CGroup: /system.slice/prometheus.service
            └─14349 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/data --web.console.

Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.787Z caller=head.go:681 level=info component=tsdb msg="On-disk WAL replay started"
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.787Z caller=head.go:689 level=info component=tsdb msg="Replay finished"
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.788Z caller=head.go:760 level=info component=tsdb msg="WAL segment loaded"
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.788Z caller=head.go:797 level=info component=tsdb msg="WAL segment loaded"
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.790Z caller=main.go:1045 level=info fs.type=EXT4_SUPER_MAGIC fs.mount=/data
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.790Z caller=main.go:1048 level=info msg="TSDB started"

```

Prometheus.service:

[Unit]

Description=Prometheus

Wants=network-online.target

After=network-online.target

StartLimitIntervalSec=500

StartLimitBurst=5

[Service]

User=prometheus

Group=prometheus

Type=simple

Restart=on-failure

RestartSec=5s

ExecStart=/usr/local/bin/prometheus \

--config.file=/etc/prometheus/prometheus.yml \

--storage.tsdb.path=/data \

--web.console.templates=/etc/prometheus/consoles \

--web.console.libraries=/etc/prometheus/console_libraries \

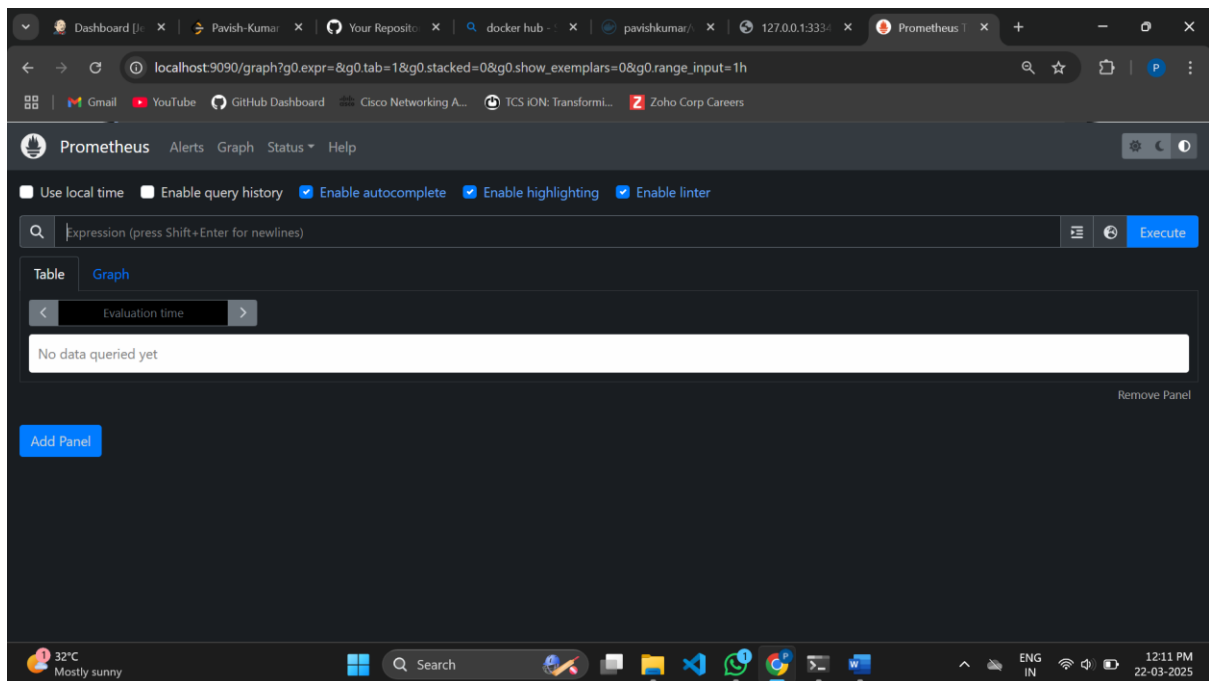
--web.listen-address=0.0.0.0:9090 \

--web.enable-lifecycle

[Install]

WantedBy=multi-user.target

```
pavis@pavish-kumar: ~  
prometheus-2.47.1.linux-amd64/prometheus  
pavis@pavish-kumar:~/prometheus-2.47.1.linux-amd64$ cd  
rm -rf prometheus-2.47.1.linux-amd64.tar.gz  
prometheus --version  
sudo vim /etc/systemd/system/prometheus.service  
prometheus, version 2.47.1 (branch: HEAD, revision: c4d1a8beff37cc004f1dc4ab9d2e73193f51aaeb)  
  build user: root@4829330363be  
  build date: 20231004-10:31:16  
  go version: go1.21.1  
  platform: linux/amd64  
  tags: netgo,builtinassets,stringlabels  
pavis@pavish-kumar:~$ sudo vim /etc/systemd/system/prometheus.service  
pavis@pavish-kumar:~$ sudo vim /etc/systemd/system/prometheus.service  
pavis@pavish-kumar:~$ sudo systemctl enable prometheus  
sudo systemctl start prometheus  
sudo systemctl status prometheus  
journalctl -u prometheus -f --no-pager  
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.  
● prometheus.service - Prometheus  
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2025-03-22 06:40:12 UTC; 21ms ago  
     Main PID: 14349 (prometheus)  
       Tasks: 1 (limit: 4592)  
      Memory: 2.4M (-)  
    CGroup: /system.slice/prometheus.service  
            └─14349 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/data --web.console.l  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.787Z caller=head.go:681 level=info component=tsdb msg="On-dis  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.787Z caller=head.go:689 level=info component=tsdb msg="Replay  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.788Z caller=head.go:760 level=info component=tsdb msg="WAL se  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.788Z caller=head.go:797 level=info component=tsdb msg="WAL re  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.790Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC  
Mar 22 06:40:12 pavish-kumar prometheus[14349]: ts=2025-03-22T06:40:12.790Z caller=main.go:1048 level=info msg="TSDB started"
```



Node Exporter:

Commands for installation:

```
sudo mv \  
node_exporter-1.6.1.linux-amd64/node_exporter \  
/usr/local/bin/
```

```
rm -rf node_exporter*
```

Node exporter file:

[Unit]

Description=Node Exporter

Wants=network-online.target

After=network-online.target

StartLimitIntervalSec=500

StartLimitBurst=5

[Service]

User=node_exporter

Group=node_exporter

Type=simple

Restart=on-failure

RestartSec=5s

ExecStart=/usr/local/bin/node_exporter \

 --collector.logind

[Install]

WantedBy=multi-user.target

Service check : sudo systemctl enable node_exporter

sudo systemctl start node_exporter

sudo systemctl status node_exporter

journalctl -u node_exporter -f --no-pager

```
$ my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label 'job=<job_name>' to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: "node_export"
    static_configs:
      - targets: ["localhost:9100"]

  - job_name: "jenkins"
    metrics_path: '/prometheus'
    static_configs:
      - targets: ["localhost:8080"]

...
...
...

"/etc/prometheus/prometheus.yml" 38L, 1132B 31,27 All
```

Reload Prometheus: `curl -X POST http://localhost:9090/-/reload`

Targets

All scrape pools: All Unhealthy Collapse All Filter by endpoint or labels Unknown Unhealthy Healthy

jenkins (0/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:8080/prometheus	DOWN	instance="localhost:8080" job="jenkins"	32.331s ago	6.208ms	server returned HTTP status 403 Forbidden

node_export (1/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9100/metrics	UP	instance="localhost:9100" job="node_export"	24.854s ago	38.140ms	

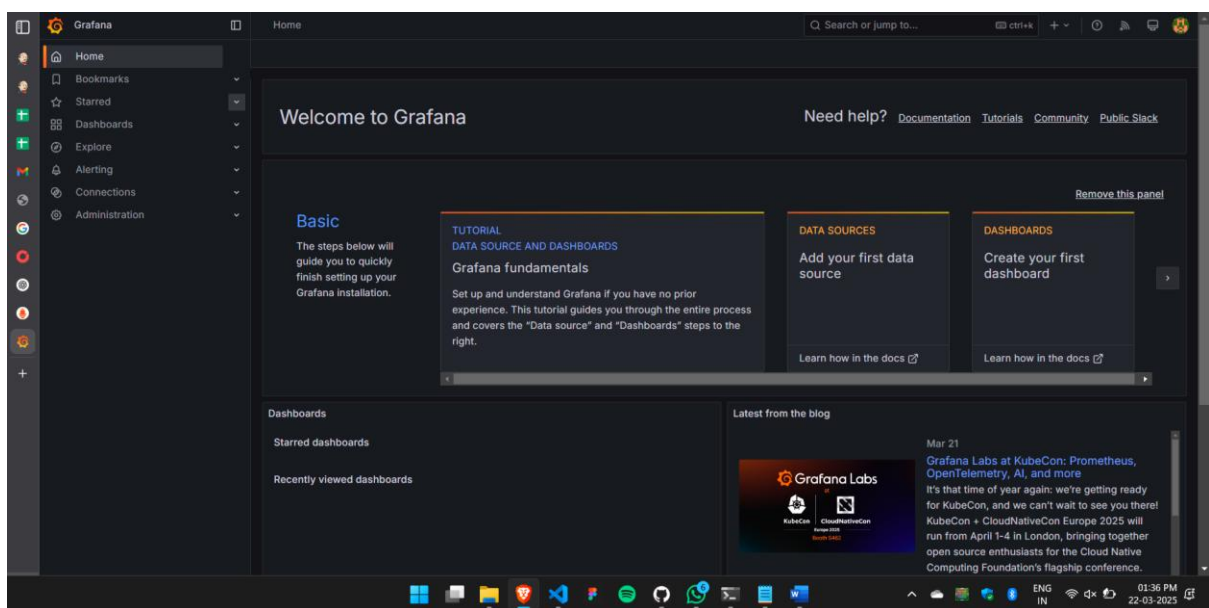
prometheus (1/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	20.766s ago	5.218ms	

Grafana:

```
sudo apt-get install -y apt-transport-https software-properties-common  
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -  
echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a  
/etc/apt/sources.list.d/grafana.list  
sudo apt-get update  
sudo apt-get -y install grafana  
sudo systemctl enable grafana-server  
sudo systemctl start grafana-server  
sudo systemctl status grafana-server
```

Grafana UI:



Status in promethues:

Targets

All Unhealthy Collapse All Filter by endpoint or labels

Unknown Unhealthy Healthy

jenkins (0/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:8080/prometheus	Down	instance="localhost:8080" job="jenkins"	-872.000ms ago	1.723ms	server returned HTTP status 403 Forbidden

node export (1/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9100/metrics	Up	instance="localhost:9100" job="node_export"	9.665s ago	22.271ms	

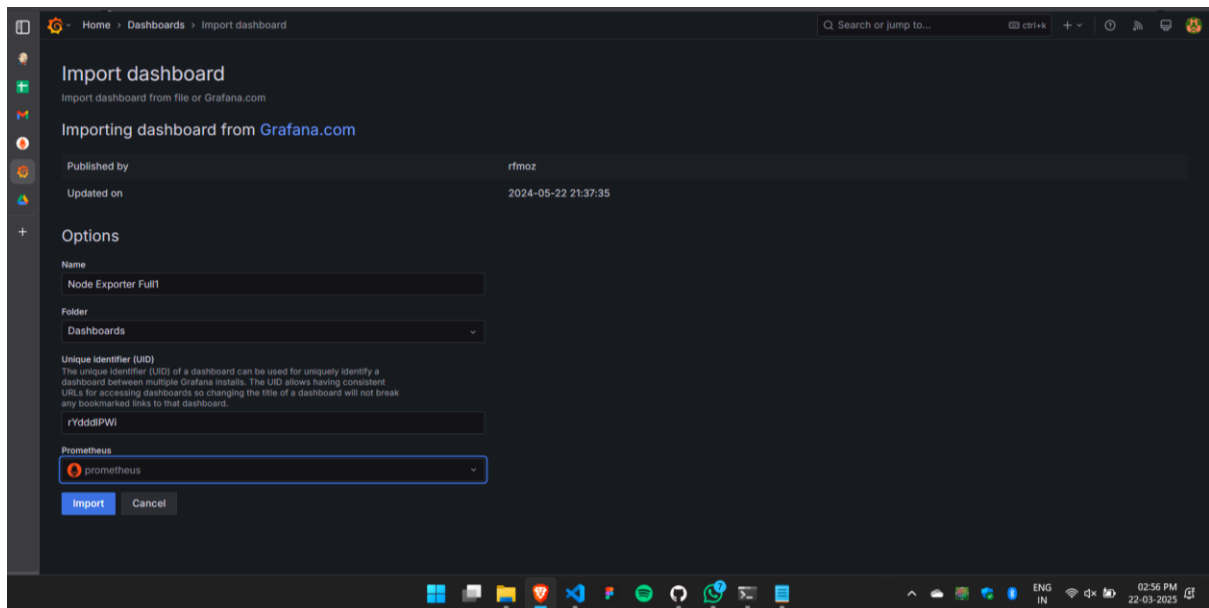
prometheus (1/1 up) [Show logs](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	Up	instance="localhost:9090" job="prometheus"	5.576s ago	3.332ms	

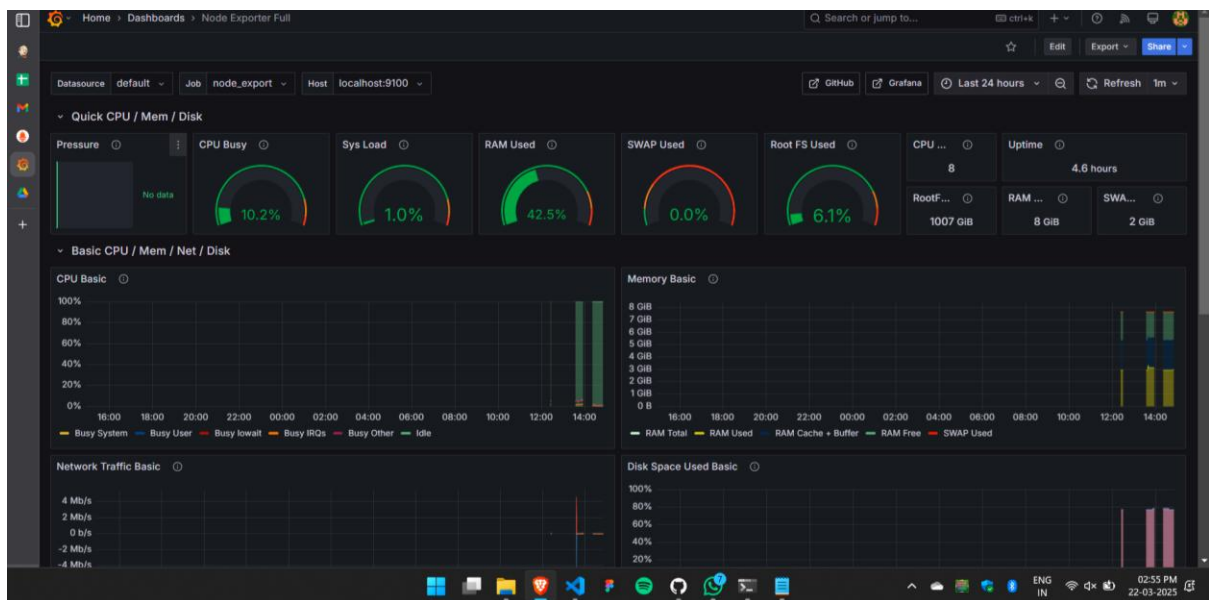
Metrics in Prometheus:

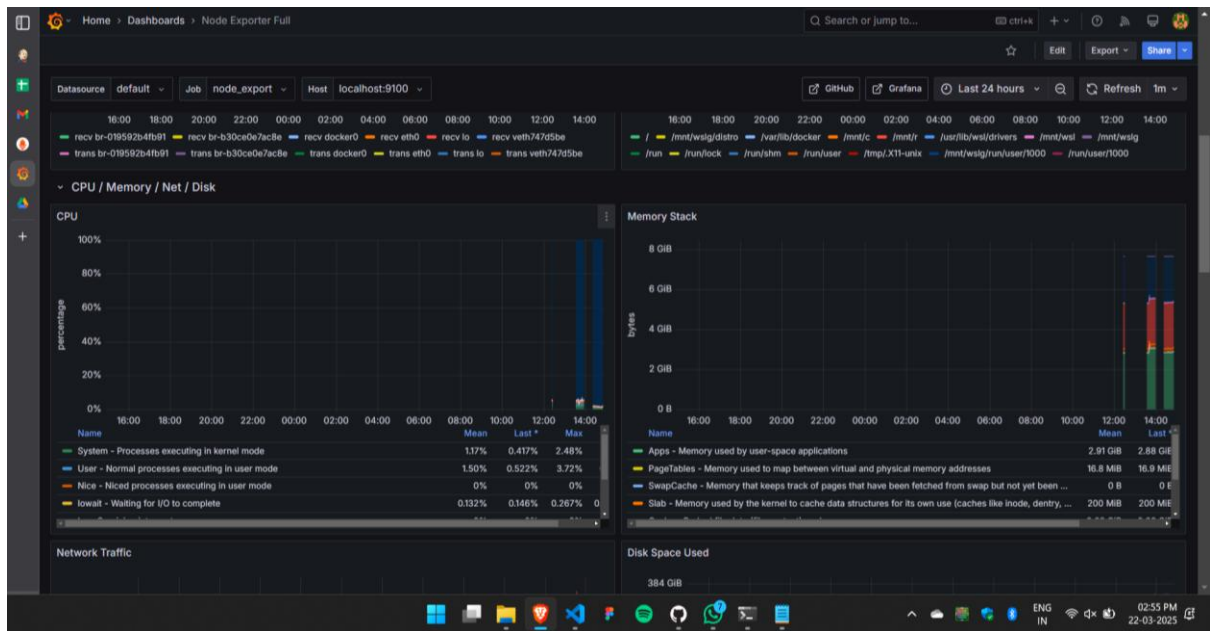
```
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 6.3341e-05
go_gc_duration_seconds{quantile="0.25"} 0.000268033
go_gc_duration_seconds{quantile="0.5"} 0.000377667
go_gc_duration_seconds{quantile="0.75"} 0.000802275
go_gc_duration_seconds{quantile="1"} 0.003268009
go_gc_duration_seconds_sum 0.027268895
go_gc_duration_seconds_count 40
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 38
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.21.1"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated and still in use.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 2.5950728e+07
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated, even if freed.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 2.8266877e+08
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.491295e+06
# HELP go_memstats_frees_total Total number of frees.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 1.446840e+06
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 4.77956e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and still in use.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 2.5950728e+07
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 2.343731e+07
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use.
# TYPE go_memstats_heap_inuse_bytes gauge
go_memstats_heap_inuse_bytes 2.5745182e+07
# HELP go_memstats_heap_objects Number of allocated objects.
# TYPE go_memstats_heap_objects gauge
go_memstats_heap_objects 118661
# HELP go_memstats_heap_released_bytes Number of heap bytes released to OS.
# TYPE go_memstats_heap_released_bytes gauge
go_memstats_heap_released_bytes 1.857945e+07
# HELP go_memstats_heap_sys_bytes Number of heap bytes obtained from system.
# TYPE go_memstats_heap_sys_bytes gauge
go_memstats_heap_sys_bytes 5.118246e+07
# HELP go_memstats_last_gc_time_seconds Number of seconds since 1970 of last garbage collection.
# TYPE go_memstats_last_gc_time_seconds gauge
go_memstats_last_gc_time_seconds 1.742634596733205e+09
# HELP go_memstats_lookups_total Total number of pointer lookups.
```

Node Exporter:



Dashboard:





Jenkins Overview:

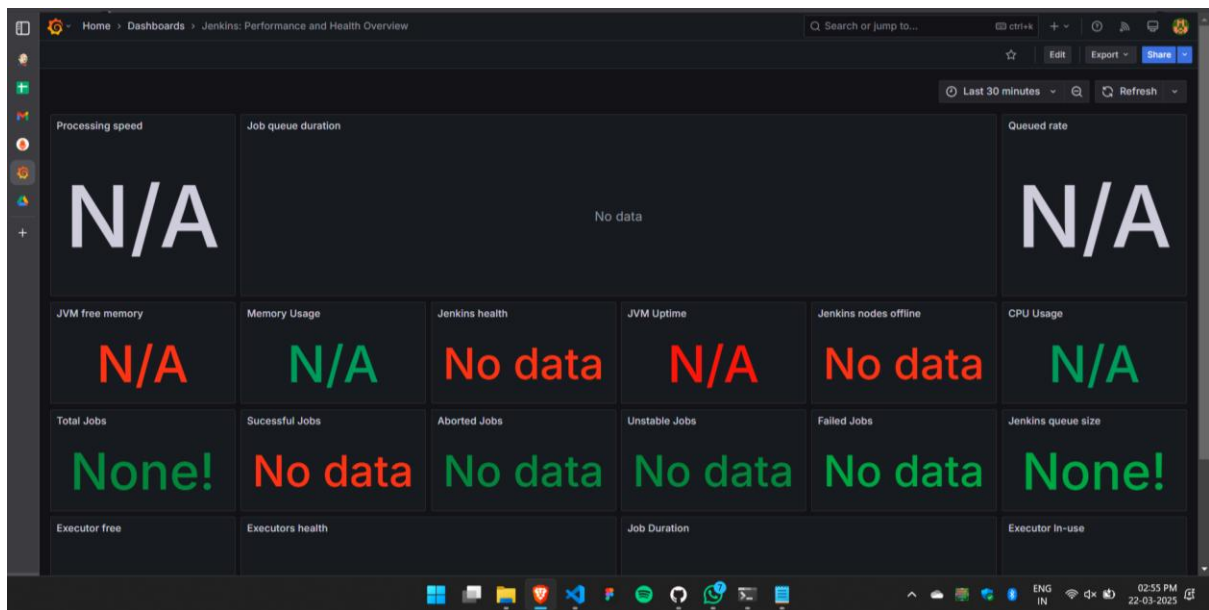
The image shows the "Import dashboard" screen in Grafana. The dashboard is titled "Jenkins: Performance and Health Overview1" and is located in the "Dashboards" folder. It was published by "haryan" and updated on "2023-08-24 15:04:53".

The "Options" section includes the following fields:

- Name: Jenkins: Performance and Health Overview1
- Folder: Dashboards
- Unique Identifier (UID): haryan-jenkin
- Prometheus: prometheus

The "Import" button is highlighted in blue.

Dashboard:

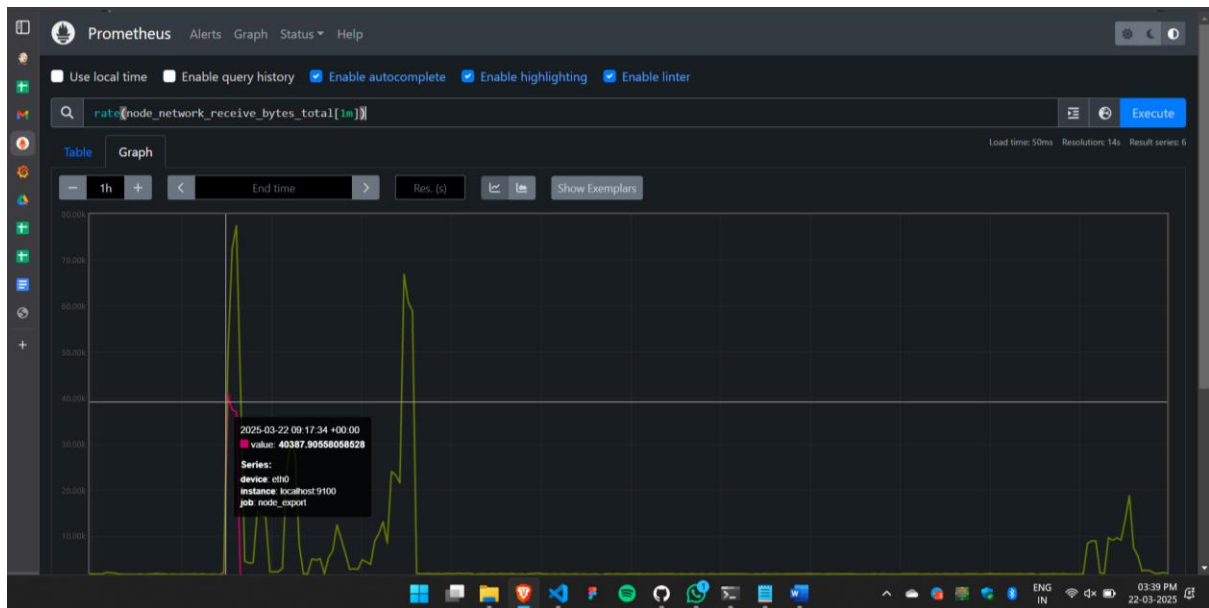


Prometheus analysis:

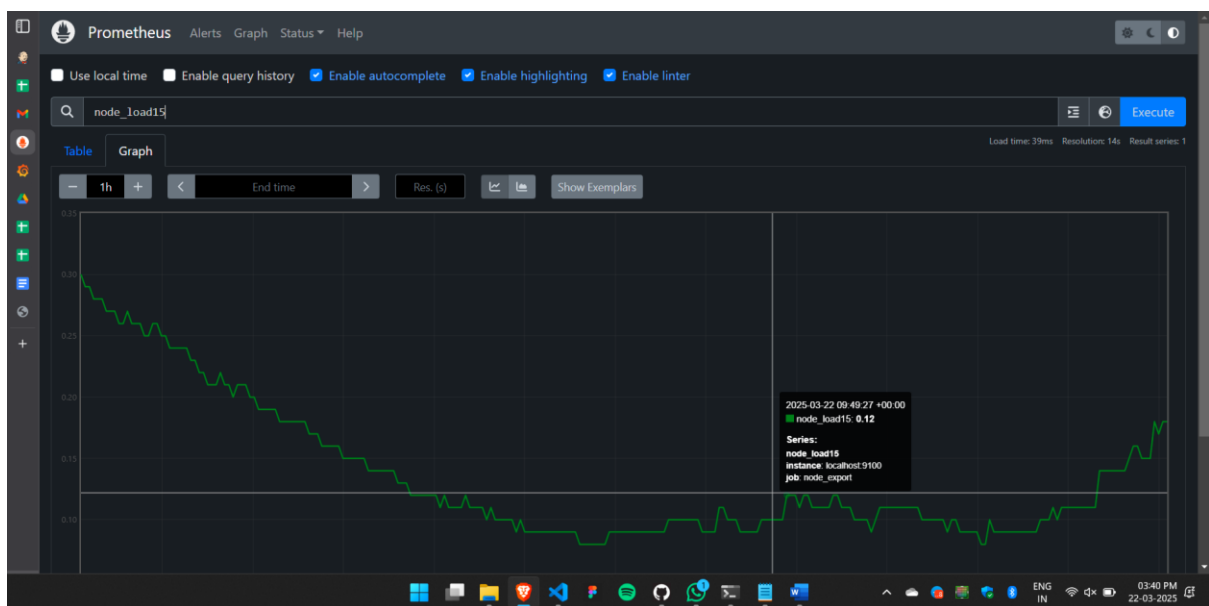
`rate(node_cpu_seconds_total{mode="system"}[1m])`



`rate(node_network_receive_bytes_total[1m])`

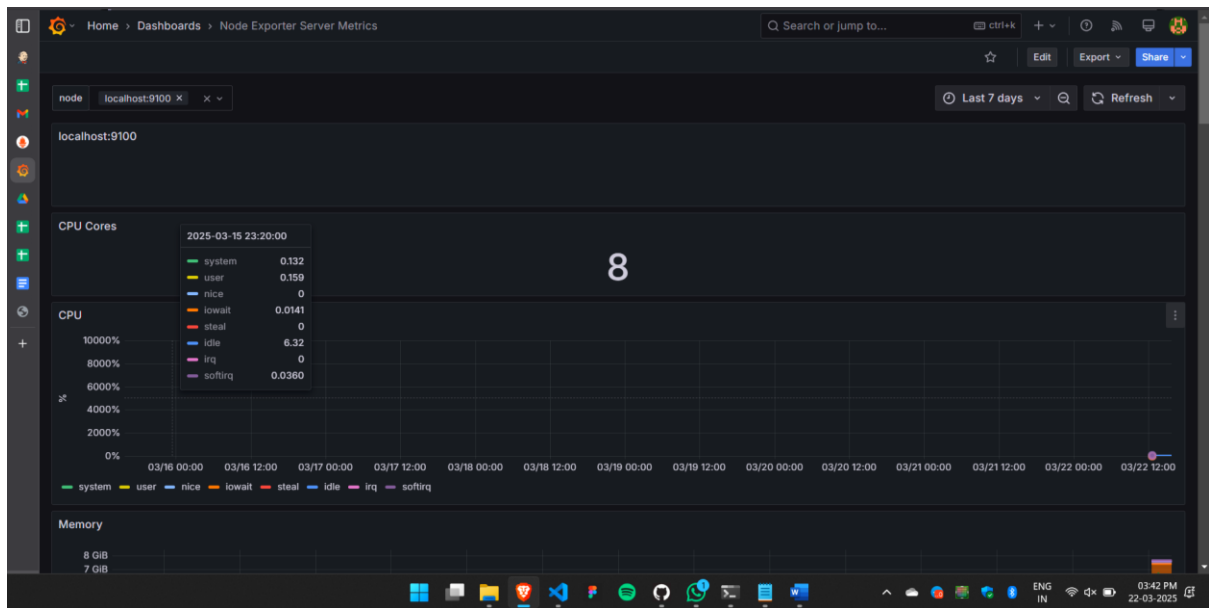


node_load15



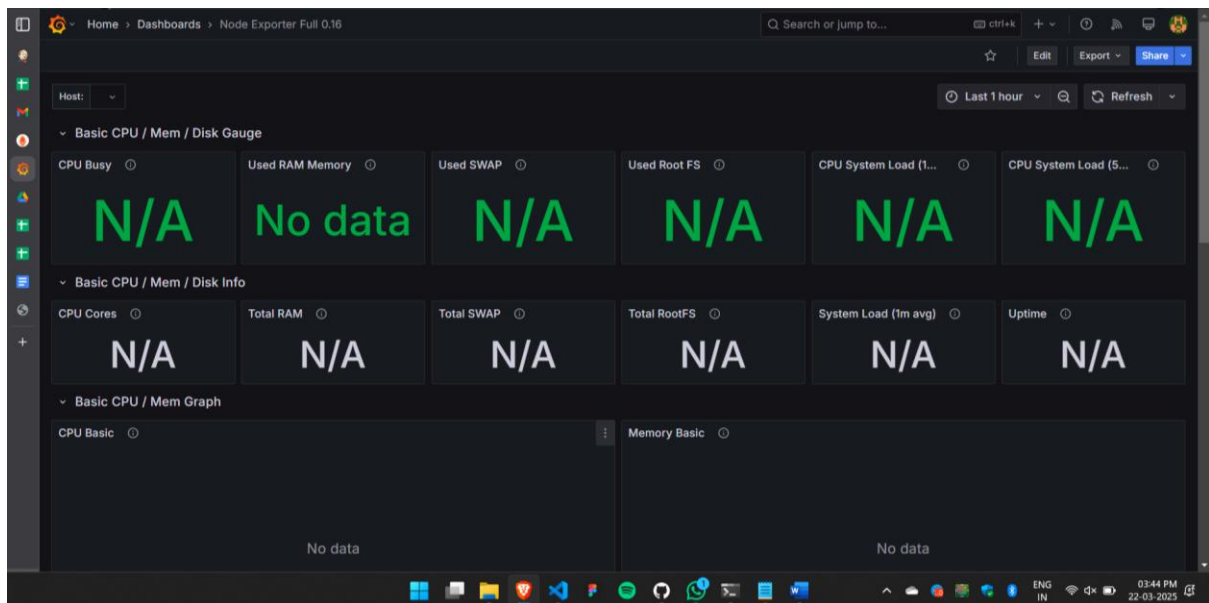
Dashboard: (405)

Node Exporter Service metrics,



Dashboard (5174):

Node Exporter Full 0.16,



Dashboard (9096):

1 Node Exporter 1.0.1

